

Characterizing Method-Level License Frictions in Open-Source Software Ecosystems

Sample adequacy, statistical design justification, and validation checks for the six-hypothesis empirical study.

1. Is the Sample Size Adequate?

A natural concern, given GitHub hosts an estimated billions of repositories while this study's samples range from **10,337 to 1,183,182**, is whether thousands-to-millions is “enough” to support ecosystem-scale claims. This conflates two distinct questions: whether an estimate is *precise* (a function of absolute sample size), and whether a sample is *representative* (a function of how it was drawn). We address both rather than treating precision as a proxy for representativeness.

1.1 Precision Depends on Absolute N, Not Population Fraction

The standard error of an estimate scales as $SE \propto 1/\sqrt{n}$, a function of absolute sample size n , not the fraction of a larger population N it represents- provided n is small relative to N , so the finite-population correction is negligible. Example: A national poll surveys ~1,000–2,000 people to represent hundreds of millions (a sampling fraction near 0.0005%) without this being considered inadequate.

By this standard every hypothesis here is comfortably powered:

Hy p.	N	Unit of Analysis	Evidence of Precision
H1	17,368	Repositories	Narrow CI on the null ($\beta=0.014$, 95% CI $[-0.014, 0.042]$), a confident null, not an uncertain one
H1a	11,252	Repositories	Estimates precise under both age specifications and a support-corrected model; the limiting factor is effect size ($AME \approx 0.1$ reusing projects), not power
H2	10,337	Projects (full census of query repos)	Bootstrap 95% CIs on all four Gini estimates are narrow (width 0.007–0.013; Section 3.4)
H3	66,136	Relationships (Category 3)	χ^2 and Cohen's h precisely estimated; effect size ($h=0.043$), not power, is the limiting factor
H4	1,116,445	Relationships (6 languages)	Smallest language group exceeds 29,000 observations; omnibus association survives repository-level aggregation (Section 3.6)
H5	868,369	Relationships (3 cohorts)	Precise at relationship level, but only 52 organizational clusters, the binding constraint is cluster count, not N (Section 3.6)

Where this study reports weak or null effects (H1, H1a, H3), the confidence intervals are narrow, not wide, evidence the null is well-estimated rather than undetectable. An underpowered study produces wide, inconclusive intervals; these are precise nulls. H5 is the one case where the effective sample size is governed by cluster count rather than observation count, and we treat it accordingly.

1.2 Representativeness Is a Separate Question, Stated Explicitly

The dataset draws from high-method-volume Corporate, OSS Foundation, and Individual cohorts rather than a uniform random sample of all GitHub repositories, and is restricted to six programming languages. This is stated explicitly as a scope boundary rather than obscured by the precision argument above: high statistical power for the sampled population does not imply the findings generalize to GitHub's long tail of low-activity repositories.

1.3 This Scope Is Standard for the Literature

No closely related study analyzes literally all of GitHub. Bissyandé et al. [2013] studied 100,000 projects; Jahanshahi et al.'s copy-based reuse work draws on World of Code, itself a large but curated index; and this study's reference database (SearchSECO) indexes ~1 million repositories and 74 million methods, already a bounded slice. “Ecosystem-scale” here denotes samples large enough to capture genuine cross-organizational and cross-language variation with statistical precision, not literal exhaustiveness.

1.4 Two Frameworks for “Enough”: Power vs. Precision

There are two ways to ask whether a sample is adequate. The **power-based** framework is the classic question asked when *planning* an experiment: given an expected effect size, how many observations are needed to detect it? It is highly sensitive to that assumed effect: at $\alpha=.05$ and 80% power, a two-group comparison needs ~785 observations for a small effect but only 49 for a large one, a sixteen-fold range driven by a quantity that can only be guessed before the study runs:

Effect Size (Cohen's d)	Required N (total, 80% power)
Small (d = 0.2)	≈ 785
Medium (d = 0.5)	≈ 126
Large (d = 0.8)	≈ 49

The **precision-based** framework asks instead: given the data available, how tight is the resulting estimate? This is measured directly via standard errors and confidence intervals, independent of any pre-specified effect size.

This study uses the precision-based framework for a straightforward reason: it analyses existing observational data rather than a purpose-designed experiment, so there was no “before the study” moment to run a power calculation against an assumed effect. What can be reported honestly is the *achieved* precision of each estimate. For this kind of study a precise, honestly reported estimate is the appropriate standard, not a power calculation requiring a guess at the very effect the data are meant to reveal.

1.5 Preliminaries: the LCD Categories and Friction Definitions

Every provenance relationship carries an integer LCD category 0–5. Two different friction definitions are used, deliberately, and the arithmetic below depends on which one applies:

Used in	Friction = 1 when category is	Friction = 0 when category is	Excluded
H2, H4	3 or 4	1, 2, or 5	0 (H2 only)
H5	3, 4, or 5	1 or 2	0

H5 uses the broader definition because it asks whether ownership predicts license-handling discipline, and unresolved metadata (Category 5) is itself evidence about governance quality rather than a neutral gap. H1/H1a do not use a friction indicator at all: their outcome is a reuse count.

2. Statistical Method Justification, Per Hypothesis

Each hypothesis uses a three-layer design: (1) a non-parametric or frequency-based screen suited to the outcome's distribution, paired with (2) a scale-free effect size that does not inflate with sample size, and (3) confound- or clustering-adjusted analysis where a plausible alternative explanation could produce a spurious result. Section 2.2's table states the method used, why it fits, and which alternatives were considered.

2.1 Why Effect Size, Not Significance Alone

A p-value answers “is this effect distinguishable from noise?” An effect size answers “how large is it?” The two diverge sharply at scale: H1's naive comparison is highly significant ($p < .001$) yet its effect size (Cliff's $\delta = 0.030$) is negligible. Unlike p-values, effect sizes do not inflate with sample size, which is why every test here is reported alongside one.

Effect Size	Used For	What It Captures
Cliff's δ	H1's naive comparison	Compares two groups' ordering: drawing one value at random from each, how often does one exceed the other, beyond chance? Ranges -1 to $+1$.
Cramér's V	H3, H4, H5 (χ^2 tests)	Strength of association between categorical variables — a correlation analogue for category data. Ranges 0 to 1.
Cohen's h	H3's goodness-of-fit test	Compares two proportions (e.g. 51.1% vs 48.9%) on a standardised, cross-study comparable scale.
IRR	H1, H1a (Negative Binomial)	Multiplicative factor: how many times more (or less) frequent the outcome is per unit change in the predictor.
ϵ^2	H4 (repository-level Kruskal-Wallis)	Share of rank variance explained by group membership; used where the unit of analysis is the repository rather than the relationship.

Each is the field-appropriate tool for its test type, and none inflate with N , a reported effect size means the same thing whether computed from 10,337 or 1.18 million observations. Effect sizes do, however, characterise association strength *in the observed corpus*; where observations are clustered they are not thereby clustering-adjusted population estimates (Section 3.6).

2.2 Method and Alternatives, Per Hypothesis

Hyp.	Method Used	Why This Method	Alternatives Considered
H1 / H1a	Mann-Whitney U (screen) → Negative Binomial regression (age-controlled)	reuse_count is right-skewed and over-dispersed (median 1, max 46–85; variance $\approx 3 \times$ mean), NegBin is standard for over-dispersed counts; Mann-Whitney needs no distributional assumptions.	PSM: addresses a different estimand (matching on a single confound) and converges to the same answer as regression adjustment here, so it lies outside this hypothesis's scope. ZINB: not applicable — the outcome contains no zeros. The relevant issue is the opposite one, zero-truncation (Section 3.2a).
H2	Gini / Lorenz curve, bootstrap CIs	H2 concerns concentration, not a group difference, Gini is the standard distribution-free inequality measure.	Atkinson/Theil: would likely agree in direction given concentration is already extreme (Gini 0.96–0.99). Theil's decomposability answers a different question (within- vs between-cohort inequality) and is noted as

Hyp.	Method Used	Why This Method	Alternatives Considered
			a future extension linking H2 and H5.
H3	χ^2 goodness-of-fit vs. 50/50, with Cohen's h	Tests whether a binary split (Permissive vs Reciprocal within Category 3) deviates from even, the standard test for this question.	The outcome (a two-category proportion) and the question (deviation from a 50/50 baseline) map directly onto this test's intended use; alternatives address different estimands.
H4	χ^2 + Cramér's V + logistic regression; repository-level aggregation	Friction is binary across six unordered languages; logistic regression adds per-language odds ratios the omnibus test cannot provide.	RF permutation importance: shifts to an exploratory ML paradigm requiring separate justification, and is one route toward the unified multivariate model already named as future work rather than a check on this result.
H5	χ^2 + Cramér's V + Holm–Bonferroni post-hoc; cluster-robust SEs; organization-level aggregation	Friction is binary across three unordered cohorts; post-hoc tests localise which pairs differ, which the omnibus result cannot.	Mixed-effects logistic regression with organization random intercepts was attempted first but failed to converge (Section 3.6); cluster-robust inference and aggregation were used instead.

Holm–Bonferroni correction is applied with an explicitly stated family. The **primary** family comprises the five hypothesis tests yielding a p-value: H1, H1a, H3, H4, H5. H2 is excluded because it is assessed through concentration measures (Gini, Lorenz, Pareto), which produce no hypothesis-test p-value. Non-primary tests are corrected within their own families: H4's per-language odds ratios, H5's three post-hoc pairwise comparisons, and the alternative model specifications used as robustness checks for H1 and H1a. Under this scheme H1a's primary test does not survive correction (raw $p = .029$, **adjusted $p = .059$**).

3. Validation and Robustness Checks

Seven independent exercises were performed to check that the pipeline's outputs are correct rather than merely internally consistent, and that the reported inferences survive the assumptions they rest on.

3.1 License Classification: Two Independent Methods, Zero Disagreements

License family (Permissive vs Reciprocal, used in H1a and H3) is classified via exact lookup against a curated SPDX mapping. This was cross-validated against a mechanically independent second method (keyword-substring matching) across the full corpus of **1,183,182** relationship rows. The two agreed on **every row**, zero disagreements, strong evidence the classification is correct rather than an artifact of either implementation.

3.2 Model Family: Negative Binomial vs. Poisson

Rather than assuming Negative Binomial is appropriate for the count outcome, a Poisson specification was fitted on the identical H1 dataset ($N = 17,368$) and compared by AIC/BIC. Negative Binomial fitted decisively better ($\Delta\text{AIC} = 5,523.98$, $\Delta\text{BIC} = 5,516.22$, against a conventional “decisive” threshold of $\Delta > 10$), consistent with the significant overdispersion parameter ($\hat{\alpha} = 0.285$, $p < .001$). This comparison establishes NB over Poisson, but both share the same untruncated support, so it does not address whether that support is correct for a positive-only outcome. Section 3.2a takes up that question.

3.3 Human Validation of the LCD Taxonomy

The six-category LCD taxonomy underlying every friction operationalisation was independently validated in prior companion work against a manually annotated ground truth of **1,934** method pairs, labelled by two researchers per SPDX compatibility guidelines (inter-rater $\kappa = 0.84$). The pipeline achieved a macro F1 of **95.0%** (precision 95.7%, recall 94.9%), with the categories most central to the actionable-debt measure, Category 3 (F1 = 97.6%) and Category 4 (100% precision), among the most reliably classified.

3.4 Bootstrap Confidence Intervals for H2's Concentration Estimates

H2's Gini coefficients were validated via non-parametric bootstrap (5,000 resamples) to confirm the concentration is a stable property of the corpus rather than of this sample:

Measure	N	Gini	95% CI
Category 3 (Procedural) volume	10,337	0.9613	[0.9545, 0.9672]
Category 4 (Structural/High Risk) volume	10,337	0.9923	[0.9901, 0.9937]
Combined actionable debt (3+4) volume	10,337	0.9604	[0.9541, 0.9659]
LCD density (debt / resolved-provenance methods)	8,407	0.7404	[0.7326, 0.7481]

All four intervals are narrow (width 0.007–0.013), confirming the reported concentration is stable and well-estimated rather than sampling noise. H2 is the only hypothesis whose headline finding survives every robustness check applied in this document.

3.5 Cross-Checking Documented Methodology Against Pipeline Behaviour

Each hypothesis's stated friction operationalisation was verified directly against its computed output on the full dataset. This confirmed that H5 intentionally uses a broader definition (Categories 3, 4 and 5) than H2 and H4 (Categories 3 and 4 only): unresolved or ambiguous licensing metadata is treated as informative about organisational governance quality for the ownership question H5 investigates, rather than as a neutral gap. This distinction is now stated explicitly in the paper's Study Design section, and every reported H5 statistic was reproduced from the governing analysis script.

3.6 Clustering: Non-Independence of Relationship-Level Observations

H3–H5 treat each provenance relationship as independent, but relationships cluster within repositories, organisations, and clone families; one widely copied method generates many rows sharing a common origin. Clustering inflates apparent precision, so naive p-values understate uncertainty. A mixed-effects logistic regression with organisation random intercepts was attempted first and **failed to converge** (posterior SDs collapsed to zero; the intercept implied a 99.8% friction rate against ~71% observed; the random-effect SD reached 143 on the logit scale). We therefore used two stable alternatives, cluster-robust standard errors and aggregation to the clustering unit, applied consistently to H4 and H5.

Check	H4 (clustered by repository)	H5 (clustered by organisation)
Cluster-robust SEs	Inflation 12–36×; only Java remains distinguishable from the C baseline ($p = .026$)	Inflation $\approx 104\times$ ($0.006 \rightarrow 0.627$); contrast no longer significant ($p = .38$)
Aggregation to cluster unit	5,362 repositories: $H = 113.25$, $p < .001$, $\epsilon^2 = 0.020$ (small)	52 organisations: $U = 247.0$, $p = .17$, Cliff's $\delta = -0.23$ (small)
Verdict	Omnibus association SURVIVES; per-language odds	Contrast does NOT survive; reported as a

Check	H4 (clustered by repository)	H5 (clustered by organisation)
	ratios are descriptive	descriptive pattern

The results differ by hypothesis, and we report them as such rather than applying a blanket caveat. H4's omnibus language association holds once repositories are the unit of analysis, though the ranking is not fully preserved: by mean repository-level friction, Java remains lowest (0.107), consistent with the relationship-level result, but C not C# is now highest (0.188), with C# falling to third (0.152). H5's cohort contrast does not: with only 52 organisational units, the study cannot distinguish a population-level ownership effect from variation across a modest number of organisations, an absence of evidence rather than evidence of absence. H3 was not re-analysed because its effect is already negligible ($h = 0.043$), and cluster-adjusted inference can only widen the interval around an effect already interpreted as absent.

3.7 Outstanding Validation Directions

Two checks remain outstanding. First, a manual sensitivity analysis resolving a random subsample of Category 4's unknown-origin cases would break the circularity currently preventing a direct procedural-versus-structural comparison in H3. Second, a larger organisational sample would allow H5's cohort contrast to be tested with adequate cluster-level power; the present analysis is limited by the number of organisations, not the number of relationships. Both are concrete next steps rather than optional extras.

Prepared as a supplementary methodological reference. All figures verified against the underlying analysis scripts and dataset; see the replication package for full reproducibility.